



AGENTIC MODEL SELECTION

AutoModel Advisor

Find the **best model** for your dataset, task, and constraints.



Best fit

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COURSE

LLMs and AI Agents for Science and Medicine

Agenda

01 **Motivation** — The model-selection problem and why it matters

02 **TransferGraph (2024)** — tool #1 for our retrieval agent

03 **ArtifactLinker (2026)** — tool #2 for our retrieval agent

04 **Our agentic approach** — How the agent finds the best models and evaluate them

05 **Evaluation** — How we measure success

06 **Course concepts demonstrated** — Techniques from the course, applied

07 **Related Work** — Existing approaches to model recommendation

MOTIVATION

Given a new ML problem with dataset, **which model should I use?**



2.8M+

models on Hugging Face alone
— and the catalog keeps growing.



one recommended model



The Problem

Choosing the best (constraints-aware) models among millions of models is slow and largely manual today.



Our Goal

Automatically recommend the best model for a given task, dataset, and real-world constraints (e.g., # model parameters).

Model selection as graph learning

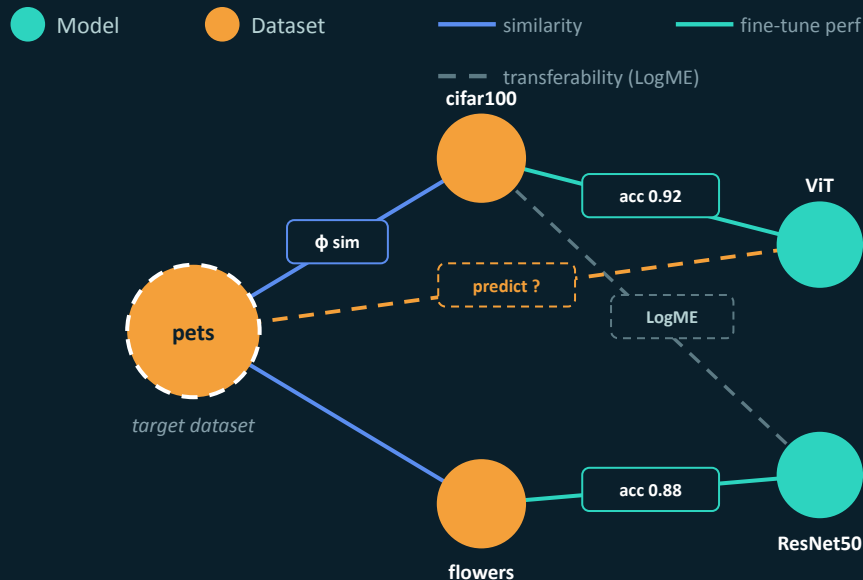
THE PROBLEM

- A model zoo holds thousands of pre-trained models. Fine-tuning all to find the best is infeasible.
- Goal: rank the models likeliest to fine-tune well on a new dataset.

REFRAMED AS

Link prediction on a model–dataset graph — score every model without fine-tuning.

Key idea: models & datasets as one graph



Learn the graph, **predict the score**

1 Collect

metadata, fine-tune history,
transferability (LogME), dataset
embeddings

2 Build & learn

a model–dataset graph;
Node2Vec embeds every node
(links + weights)

3 Train predictor

LR / RF / XGBoost regress on
the training history

4 Recommend

predict fine-tune accuracy for
the target dataset → rank
models

LEAVE-ONE-OUT · INDUCTIVE

The target dataset's model edges are removed during training
— TransferGraph scores it like a brand-new node, keeping only
its dataset-similarity edges (cold-start)

RESULTS

+32%

higher correlation with real
fine-tuning vs. SOTA

TransferGraph vs LogME · Pearson correlation

image



0.37

text



0.58

Automatic SOTA discovery

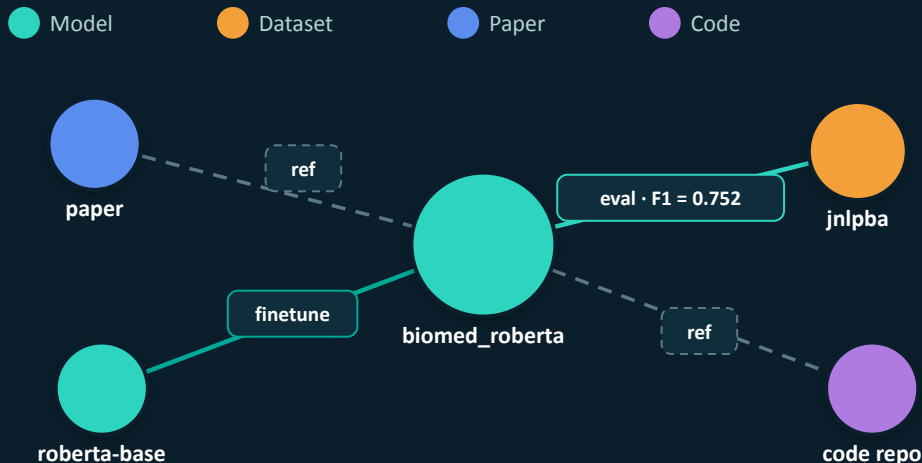
THE PROBLEM

- Hugging Face hosts millions of models and datasets.
- Evaluating every model–dataset pair is infeasible.
- Goal: automatically find pairs likely to reach a new state of the art.

REFRAMED AS

Predict promising missing links — then verify them automatically.

Key idea: Hugging Face as an artifact graph



Rank, then verify

1 Rank

A GNN scores every unobserved model–dataset pair — without running any code.

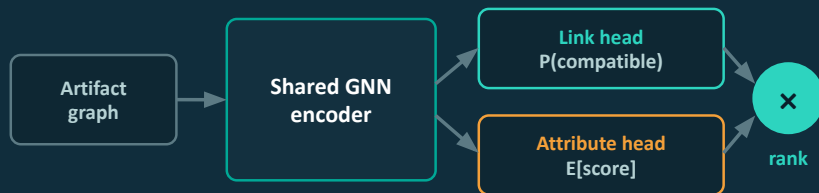
$$\text{rank} = \text{link} \times \text{attribute}$$



2 Verify

LLM coding agents **run the real evaluation and reproduce the benchmark** — only for the top-ranked candidates.

Inside the ranker: one shared GNN encoder, two heads — trained together

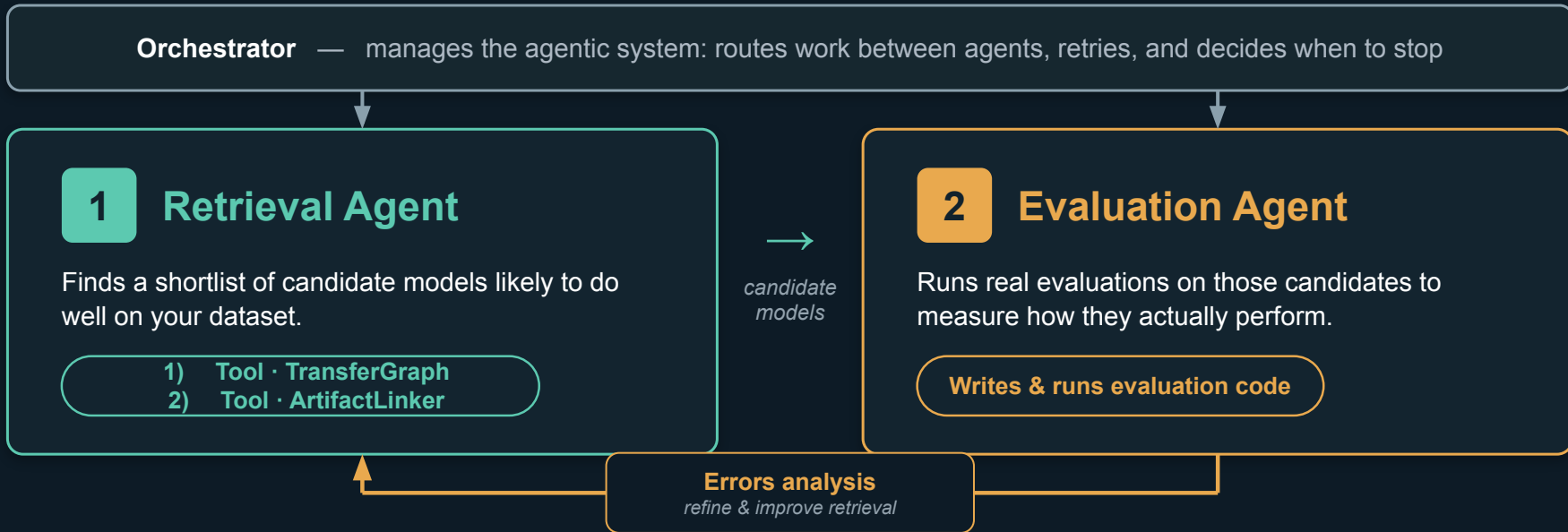


One encoder embeds each node from its neighbours; **two heads read the same embedding**.

Trained jointly: the link head (trained on negative pairs) teaches the score head which pairs are even viable — something it could never learn from observed scores alone.

Two agents: **retrieve**, then **evaluate**

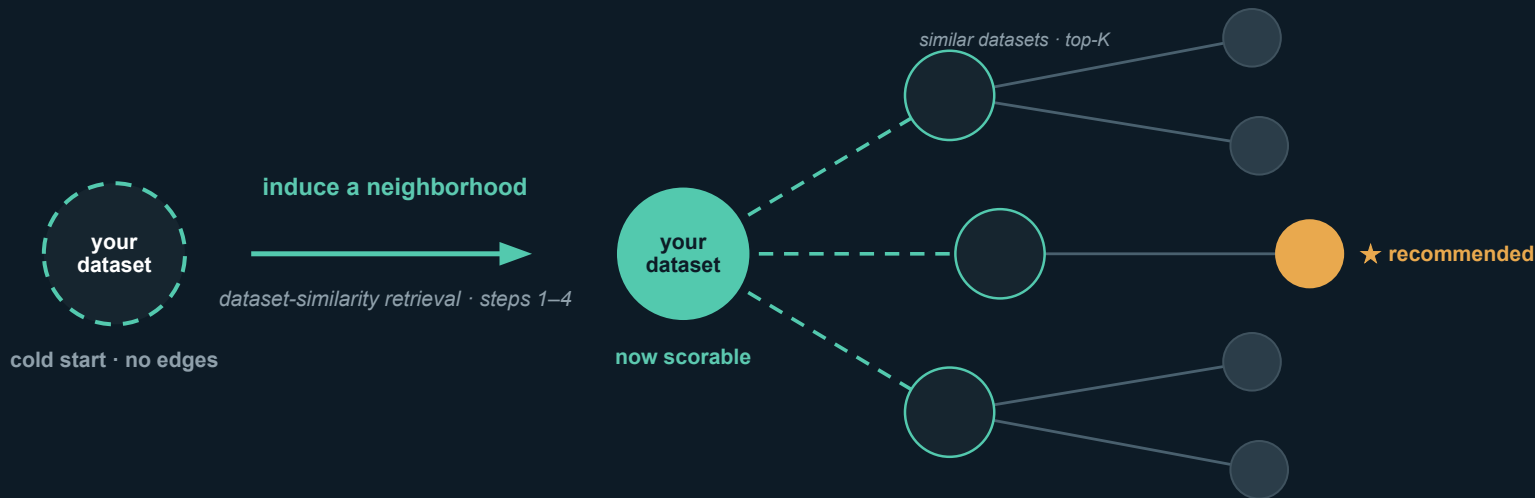
AutoModel Advisor splits model selection across two cooperating agents.



Retrieval predicts what's promising; evaluation proves it on your data.

Give a cold-start dataset a neighborhood

A brand-new dataset is a node with no edges — so no graph method can score it.



COLD-START MODEL SELECTION

Your dataset isn't in the graph — but datasets like it are

So we lend it their neighborhood — a cold-start dataset becomes scorable by an inductive recommender.

1

Your dataset

the user's data



2

LLM summary

summarize it as text



3

**Voyage
embedding**

embed the summary



4

Cosine retrieval

vs all HF dataset
embeddings → top-K



5

**LLM
recommends**

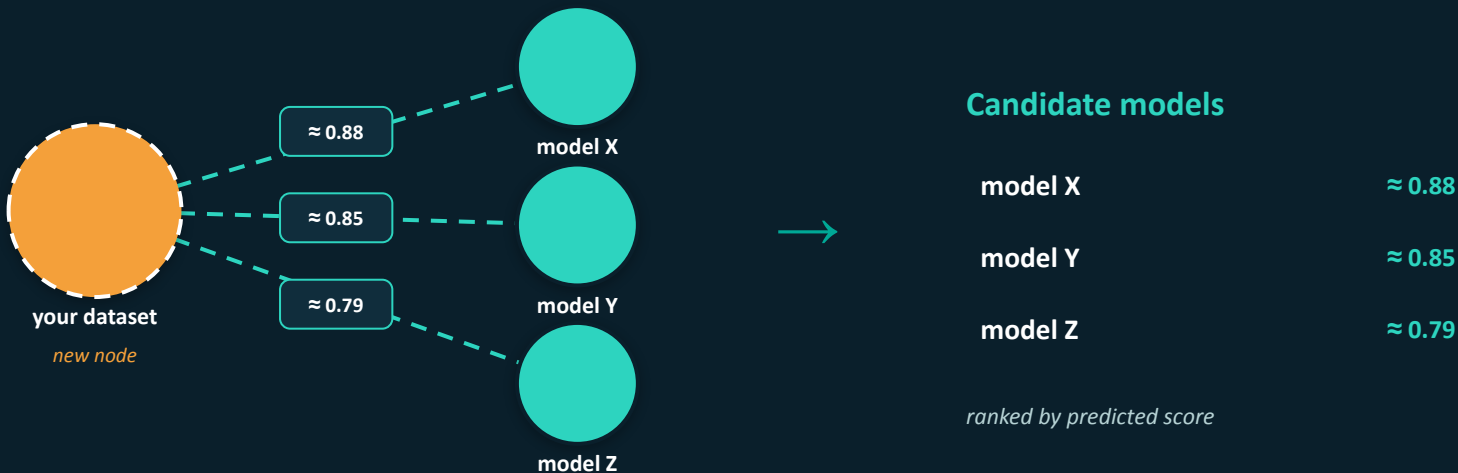
graph-conditioned
prompting

RETRIEVAL AGENT · APPROACH 2 - GNN directly

Predict directly on the graph

Same summary + embedding — but now your dataset enters the graph as a new node, and ArtifactLinker's GNN scores every model for it.

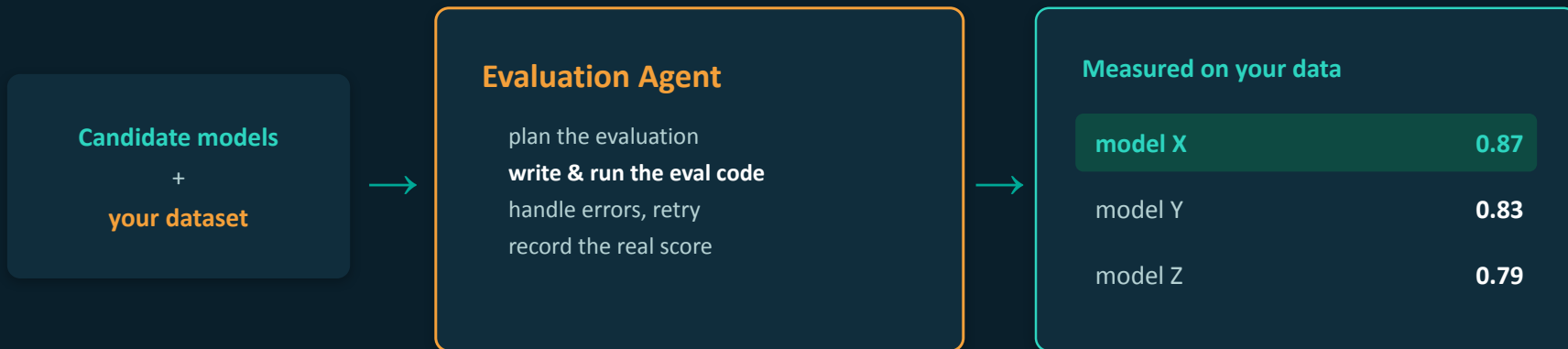
your dataset → LLM summary → Voyage embedding → **new node in the graph**



predicted score = **link** × **attribute** (ArtifactLinker's GNN). Its **inductive setting** — your dataset is a node the GNN never saw in training.

Measure performance **de-facto**

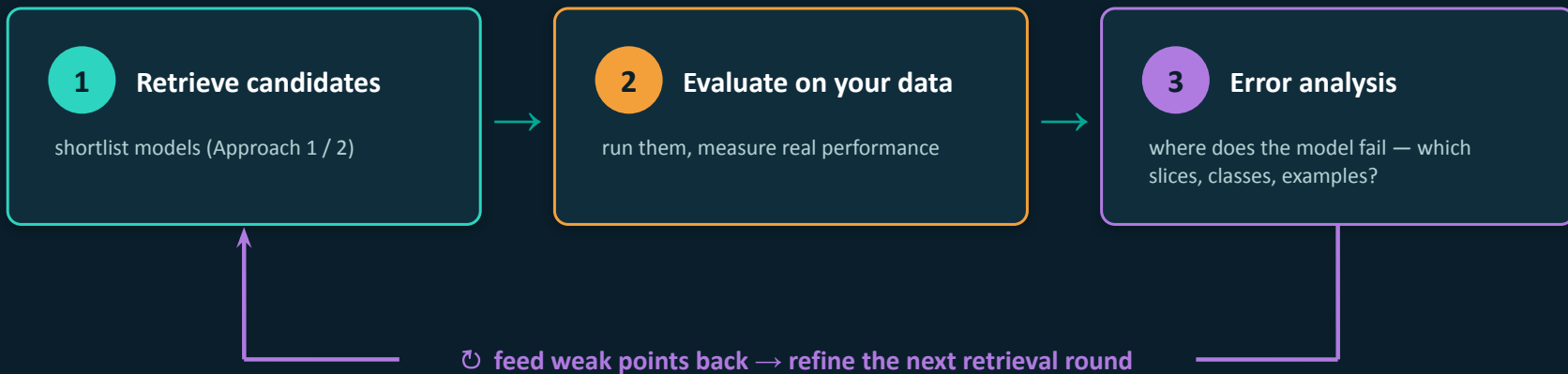
The shortlist is a prediction — the evaluation agent verifies it for real on your dataset.



We don't trust predicted scores — **the agent runs the candidates on your data and ranks by what actually happens.**

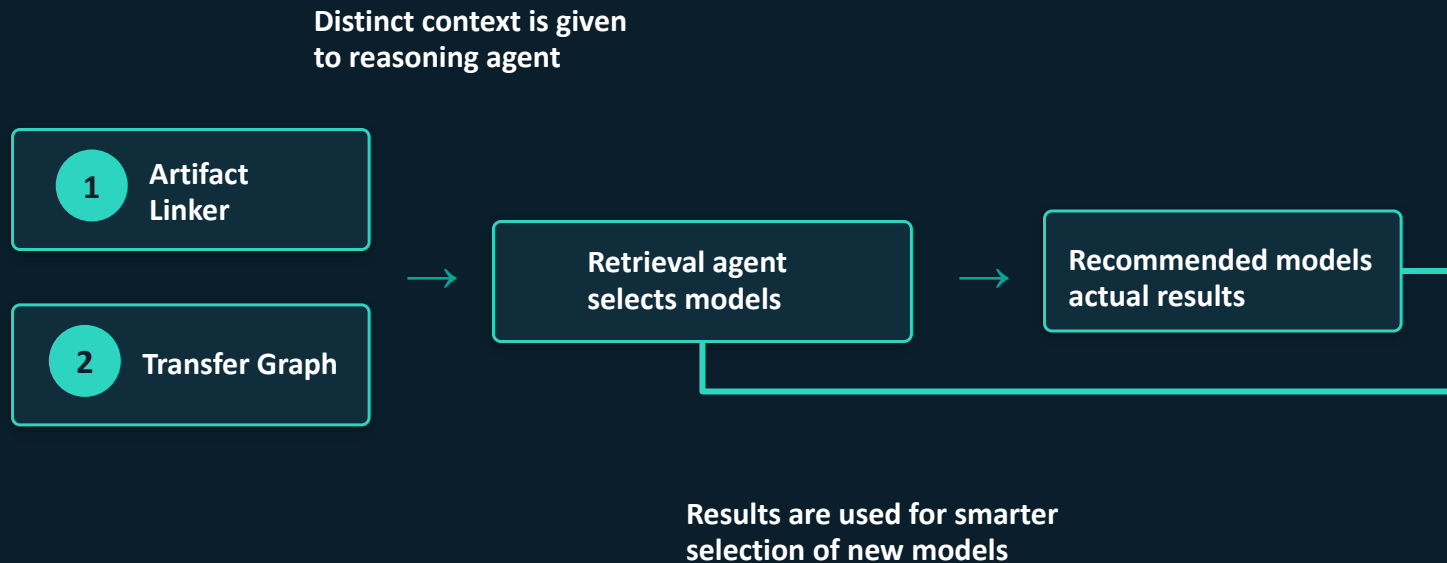
Closed-loop experimentation

After evaluation, the agent analyses where the model fails — and uses those weak points to retrieve better candidates next round.



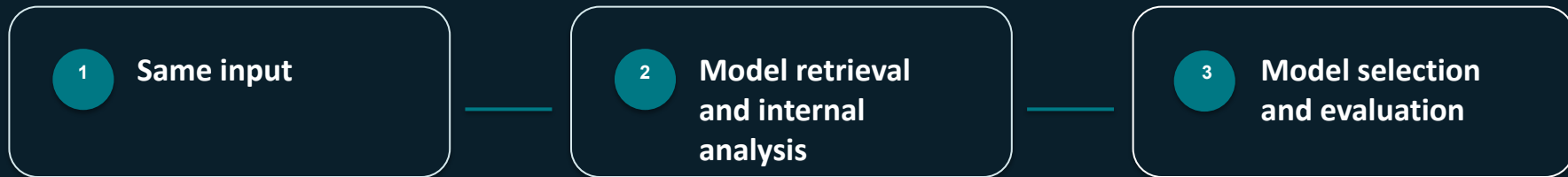
Each round targets the last round's failures — model selection becomes experimentation that improves itself, instead of a single pass.

2 types of context



Evaluation using a strong coding agent

Question: can a coding agent solve model selection by search + trial runs alone?



Important control:

Same pilot subset, evaluation protocol, hardware, constraints, and computing budget.

Techniques and Ideas from the course, **applied**



Multi-agent workflow

Retrieval agent shortlists; evaluation agent runs the candidates.



Tool use

ArtifactLinke, transfer graph, Voyage embeddings, HF Hub, code sandbox.



Data Insights

Decisions grounded in graph context and dataset information.



Plan & execute experiments

Runs the shortlist on your data to measure real performance.



Closed-loop experimentation

Analyse failures → refine retrieval → repeat.

From a user's dataset to a recommendation that has been tested - not just predicted.

RELATED WORK

Pre-trained model selection

LogME (2021) [1]

Model Spider (2023) [2]

LLM / agent-based discovery

HuggingGPT (2023)[3]

AutoML-Agent (2024) [4]

Closed-loop experimentation

DS-Agent (2024) [5]

R&D-Agent (2025) [6]

[1] You et al., “LogME: Practical Assessment of Pre-trained Models for Transfer Learning,” 2021.

[2] Zhang et al., “Model Spider: Learning to Rank Pre-Trained Models Efficiently,” 2023.

[3] Shen et al., “HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face,” 2023.

[4] Trirat et al., “AutoML-Agent: A Multi-Agent LLM Framework for Full-Pipeline AutoML,” 2024.

[5] Guo et al., “DS-Agent: Automated Data Science by Empowering LLMs with Case-Based Reasoning,” 2024.

[6] Yang et al., “R&D-Agent: Automating Data-Driven AI Solution Building Through LLM-Powered Automated Research, Development, and Evolution,” 2025.

Our contribution

AutoModel Advisor builds on the graph-based and agentic streams - **And uses them to produce a better product.**

1

Agentic: Closed-loop model selection and evaluation

Pilot results and error analysis guide the next retrieval round - each evaluation changes which models the agent tests next.

2

Built around real constraints

Optimizes for the user's dataset and practical limits, so the final model is feasible and cost-effective - not only high-scoring.